CN Lab 2 - Subnetting Assignment Report

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Question 1:

You are given the network:
192.168.10.0/25

Your task is to divide the network into valid subnets and determine the host ranges. Fill the following table.

Answer:

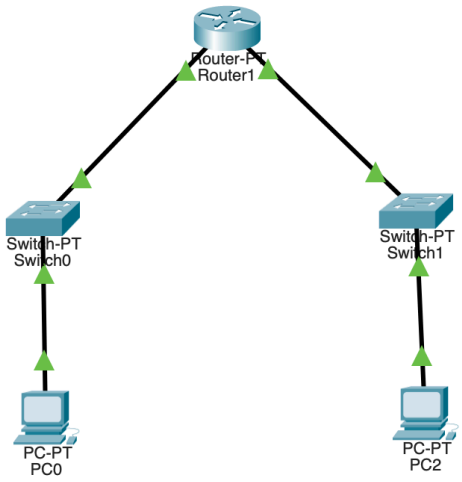
Subnet	Network Address	First Host	Last Host	Broadcast Address
Subnet 1	192.168.10.0	192.168.10.1	192.168.10.126	192.168.10.127
Subnet 2	192.168.10.128	192.168.10.129	192.168.10.254	192.168.10.255

Question 2:

Using Cisco Packet Tracer:

1. **Create a simple network containing:**
    - **One router**
    - **Two switches**
    - **Two PCs**
2. **Assign IP addresses from the two subnets created earlier.**

Answer:



PC IP Configuration:

PC	IP Address	Subnet Mask	Default Gateway
PC1	198.168.10.2	255.255.255.128	198.168.10.1
PC2	198.168.10.130	255.255.255.128	198.168.10.129

Router Interface Configuration:

```
enable
configure terminal

! Subnet 1 interface
interface f0/0
ip address 198.168.10.1 255.255.255.128
no shutdown
exit

! Subnet 2 interface
Interface f 1/0
ip address 198.168.10.129 255.255.255.128
no shutdown
exit
```

end
write memory

Question 3:

Configure the router interface as the default gateway for both networks.

Answer:

- PC1 default gateway = `192.168.10.1`
- PC2 default gateway = `192.168.10.129`

(Already configured in the table above and router interfaces.)

Question 4:

Ping between PC1 and PC2 and record the result.

Answer:

```
C:\>ping 198.168.10.130

Pinging 198.168.10.130 with 32 bytes of data:

Request timed out.
Reply from 198.168.10.130: bytes=32 time<1ms TTL=127
Reply from 198.168.10.130: bytes=32 time<1ms TTL=127
Reply from 198.168.10.130: bytes=32 time<1ms TTL=127

Ping statistics for 198.168.10.130:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Advanced Topology Question:

Design the IP addressing for the topology shown with 4 PCs and 2 routers (each LAN with 25 hosts).

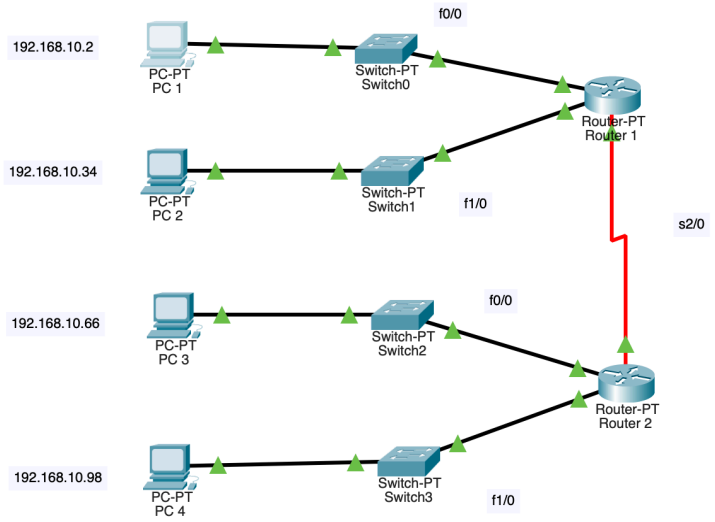
Answer:

Subnetting for 25-host networks (/27)

Subnet	Network Address	Host Range	Broadcast Address
Subnet 1	192.168.10.0	192.168.10.1 – 192.168.10.30	192.168.10.31
Subnet 2	192.168.10.32	192.168.10.33 – 192.168.10.62	192.168.10.63
Subnet 3	192.168.10.64	192.168.10.65 – 192.168.10.94	192.168.10.95
Subnet 4	192.168.10.96	192.168.10.97 – 192.168.10.126	192.168.10.127

PC IP Configuration (Advanced Topology)

PC	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.2	255.255.255.224	192.168.10.1
PC2	192.168.10.34	255.255.255.224	192.168.10.33
PC3	192.168.10.66	255.255.255.224	192.168.10.65
PC4	192.168.10.98	255.255.255.224	192.168.10.97



Router R1 Configuration

```
enable
configure terminal

interface f0/0
ip address 192.168.10.1 255.255.255.224
no shutdown
exit

interface f1/0
ip address 192.168.10.33 255.255.255.224
no shutdown
exit

interface s2/0
ip address 192.168.10.161 255.255.255.252
no shutdown
exit

! Static routes to R2
ip route 192.168.10.64 255.255.255.224 192.168.10.162
ip route 192.168.10.96 255.255.255.224 192.168.10.162

end
write memory
```

Router R2 Configuration

```
enable
configure terminal

interface f0/0
ip address 192.168.10.65 255.255.255.224
no shutdown
exit

interface f1/0
ip address 192.168.10.97 255.255.255.224
no shutdown
exit

interface s2/0
ip address 192.168.10.162 255.255.255.252
no shutdown
exit
```

! Static routes to R1
ip route 192.168.10.0 255.255.255.224 192.168.10.161
ip route 192.168.10.32 255.255.255.224 192.168.10.161

end
write memory

Test Connectivity:

From PC1:

ping 192.168.10.98

```
C:\>ping 192.168.10.98

Pinging 192.168.10.98 with 32 bytes of data:

Reply from 192.168.10.98: bytes=32 time=8ms TTL=126
Reply from 192.168.10.98: bytes=32 time=17ms TTL=126
Reply from 192.168.10.98: bytes=32 time=44ms TTL=126
Reply from 192.168.10.98: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.10.98:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 44ms, Average = 19ms

C:\>
```